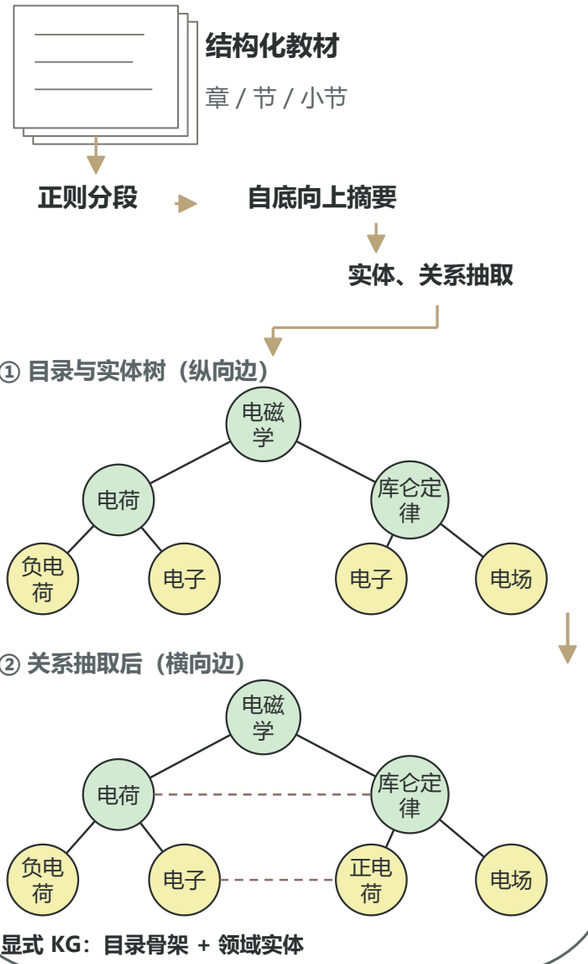
Tree-KG｜从教材树形骨架到可扩展知识图谱

依据 Niu et al., ACL 2025, Figure 2 与 §3.2–3.3 的可编辑讲解版

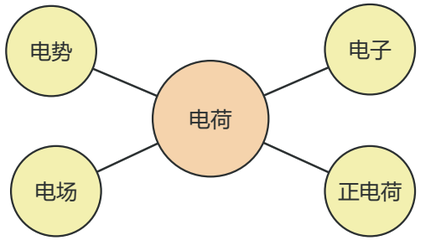
I 初始构建 • Explicit KG



显式 KG：目录骨架 + 领域实体

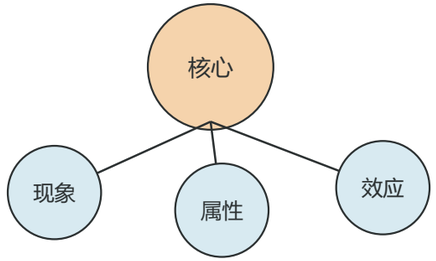
II 迭代扩展 • Hidden KG

1 上下文卷积



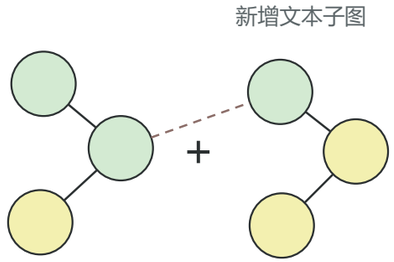
邻域语义 → 节点描述

2 实体聚合



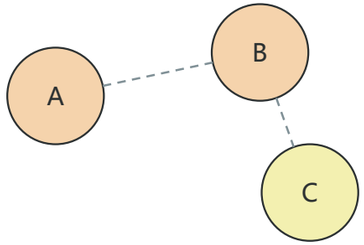
横向关联 → 核心/非核心纵向边

6 结构整合



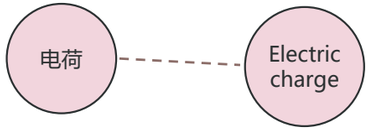
新文本子图 → 对齐并合并

3 节点嵌入



描述 → 向量；近邻检索

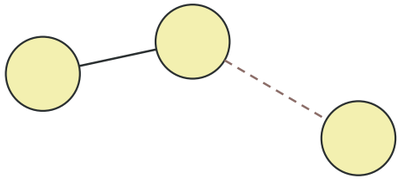
4 实体去重



同角色 + 近邻 + LLM 判同

并查集合并

5 边预测



$\cos(z_u, z_v)$ + AA + CA

排序 → LLM 强度阈值 → 加边

扩展后的 Tree-KG

图注：六类算子可灵活组合；图中不规定固定执行顺序。颜色区分类别；虚线表示横向、候选或推断关系。